

CO2

ASSESSMENT TOOL-2

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DATE:23.07.2026

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Thursday

COURSE CODE : CSA1103

COURSE NAME : OOAD

1. Use Case Diagram Analysis – Smart Hotel Reservation System

Problem Analysis

The Smart Hotel Reservation System allows guests to search for rooms, reserve rooms, cancel reservations, and make payments. Hotel managers manage room allocation and generate reports. An external payment service processes online transactions.

The existing design does not clearly separate the responsibilities of each actor. Therefore, additional actors and use cases are required.

a) Actors

Primary Actors

1. **Guest**

1. Searches available rooms
2. Makes reservations
3. Cancels reservations
4. Makes payments
5. Views booking details

2. **Hotel Manager**

1. Manages room allocation
2. Updates room availability
3. Generates reports

3. **Receptionist**

1. Checks guest reservations
2. Handles check-in and check-out
3. Updates booking status

Supporting Actors

1. **Payment Service**

1. Processes payments
2. Verifies transactions
3. Processes refunds

2. **Notification Service**

1. Sends booking confirmation
2. Sends cancellation notification
3. Sends payment confirmation

b) Primary Use Cases

- Search Rooms
- Check Room Availability
- View Room Details
- Make Reservation
- Confirm Reservation
- Cancel Reservation
- Process Payment
- Verify Payment
- Process Refund
- Allocate Room
- Manage Room Availability
- Check-in Guest
- Check-out Guest
- Generate Reports
- Send Notification

c) Relationships

<<include>> Relationships

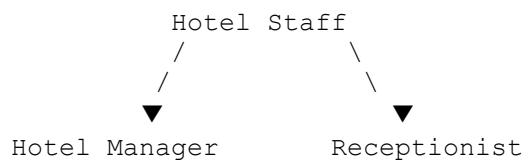
1. Search Rooms <<include>> Check Room Availability
2. Make Reservation <<include>> Check Room Availability
3. Make Reservation <<include>> Process Payment
4. Process Payment <<include>> Verify Payment
5. Make Reservation <<include>> Send Confirmation
6. Cancel Reservation <<include>> Process Refund

<<extend>> Relationships

1. Cancel Reservation <<extend>> Manage Reservation
2. Apply Discount <<extend>> Make Reservation
3. Send Cancellation Notification <<extend>> Cancel Reservation

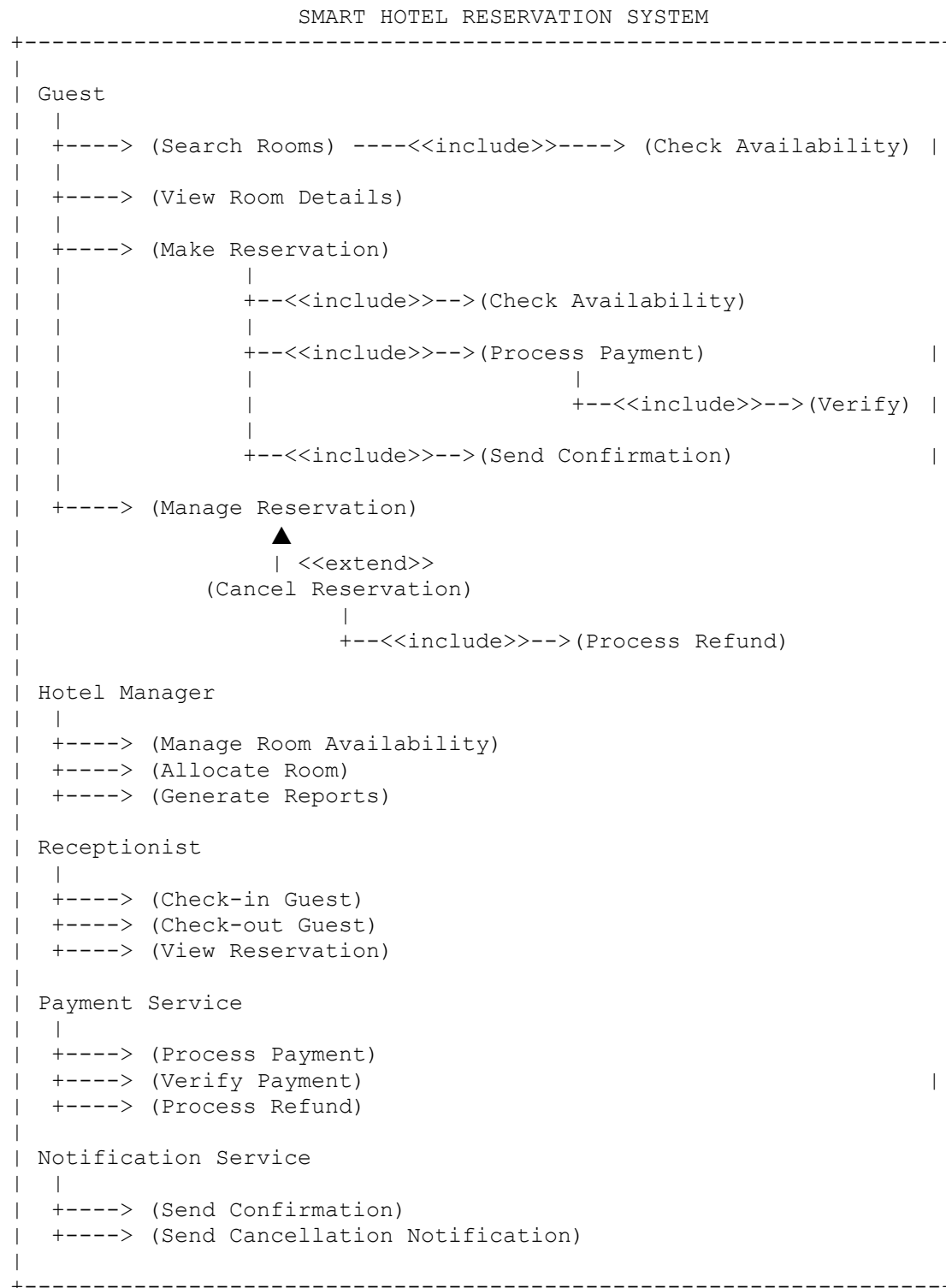
Generalization

A generalized actor **Hotel Staff** can be created.



Both Hotel Manager and Receptionist are specialized types of Hotel Staff.

d) Optimized Use Case Diagram



Conclusion

The optimized Use Case Diagram clearly separates the responsibilities of guests, hotel staff, payment services, and notification services. `<<include>>` is used for mandatory reusable activities, `<<extend>>` for optional or conditional behavior, and generalization is used to represent common hotel staff responsibilities.

2. Class Diagram Design – University Management System

Problem Analysis

The University Management System manages students, faculty, courses, departments, and examinations. The classes must have appropriate attributes, operations, and relationships.

a) Classes, Attributes and Methods

1. Student

Attributes:

- studentId
- name
- email
- phone
- dateOfBirth

Methods:

- registerCourse()
- attendExam()
- viewResult()

2. Faculty

Attributes:

- facultyId
- name
- email
- specialization

Methods:

- teachCourse()
- conductExam()
- enterMarks()

3. Course

Attributes:

- courseId
- courseName
- credits
- duration

Methods:

- addCourse()
- updateCourse()
- assignFaculty()

4. Department

Attributes:

- departmentId
- departmentName
- location

Methods:

- addCourse()
- assignFaculty()
- manageStudents()

5. Examination

Attributes:

- examId
- examName
- examDate
- examTime
- maximumMarks

Methods:

- scheduleExam()
- conductExam()
- publishResult()

6. Person – Generalized Class

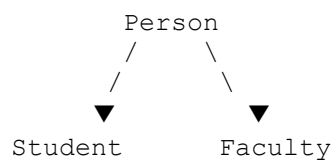
Attributes:

- name
- email
- phone

Methods:

- login()
- updateProfile()

Inheritance



b) Relationships

Association

- Student enrolls in Course.
- Faculty teaches Course.
- Student takes Examination.
- Faculty conducts Examination.

Aggregation

- Department ◇—— Course
- Department ◇—— Faculty
- Department ◇—— Student

A department contains students, faculty members, and courses.

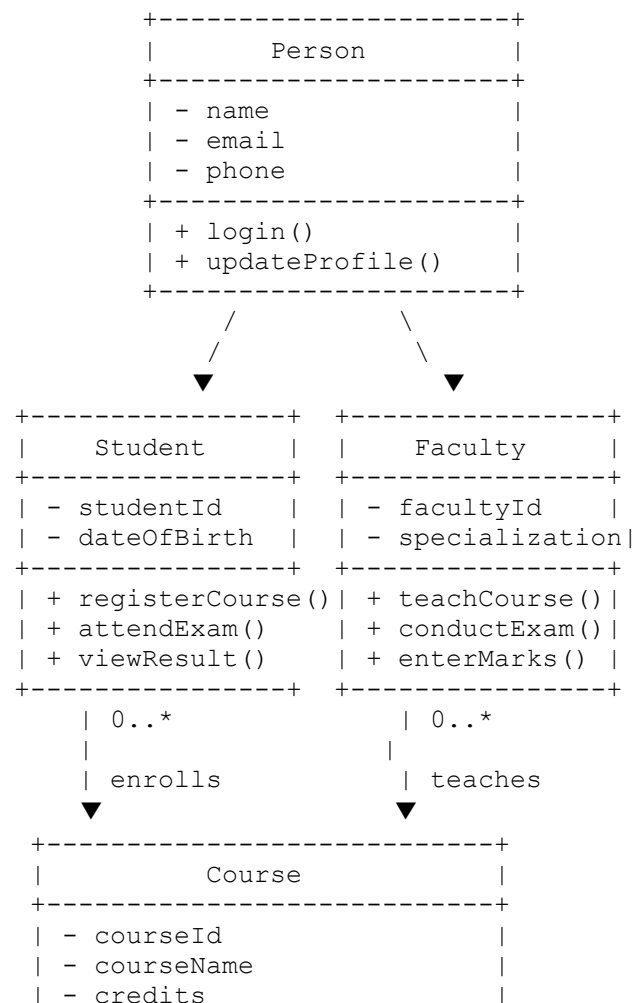
Inheritance

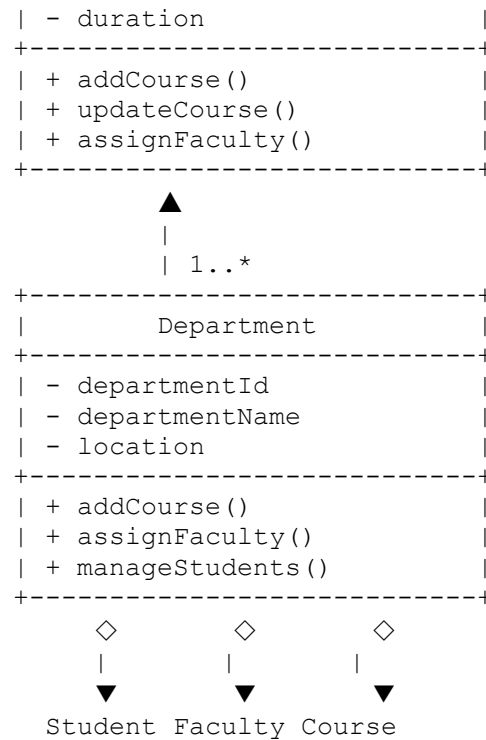
- Student inherits from Person.
- Faculty inherits from Person.

c) Multiplicity

Relationship	Multiplicity
Department - Student	1 : 0..*
Department - Faculty	1 : 1..*
Department - Course	1 : 1..*
Student - Course	0..* : 0..*
Faculty - Course	1 : 0..*
Student - Examination	1 : 0..*
Faculty - Examination	1 : 0..*

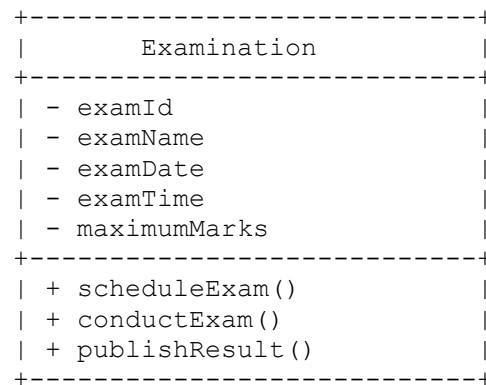
d) Class Diagram





Student 0..* ----- 0..* Examination
attends

Faculty 1 ----- 0..* Examination
conducts



Conclusion

The Class Diagram provides a well-structured representation of the University Management System. It uses **association** for interaction between classes, **aggregation** for Department ownership relationships, and **inheritance** to represent common properties of Student and Faculty.

3. Sequence Diagram Evaluation – Vehicle Rental System

Problem Analysis

The Vehicle Rental System requires interaction between the customer, rental application, vehicle database, booking system, payment gateway, and notification service.

The missing objects in the incomplete design are:

- Customer
- Rental System
- Vehicle Database
- Booking System
- Payment Gateway
- Notification Service

a) Objects

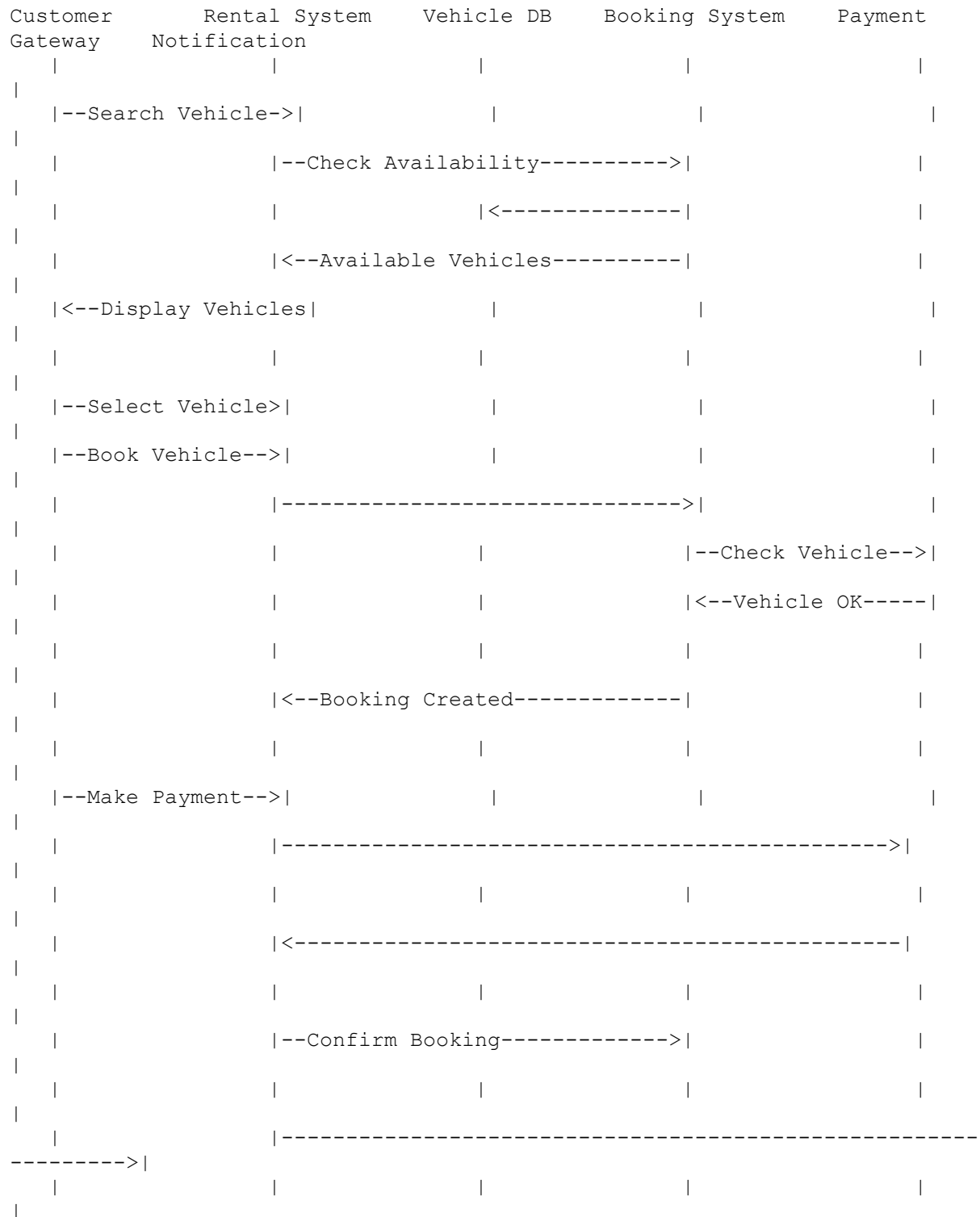
1. Customer
2. Rental System
3. Vehicle Database
4. Booking System
5. Payment Gateway
6. Notification Service

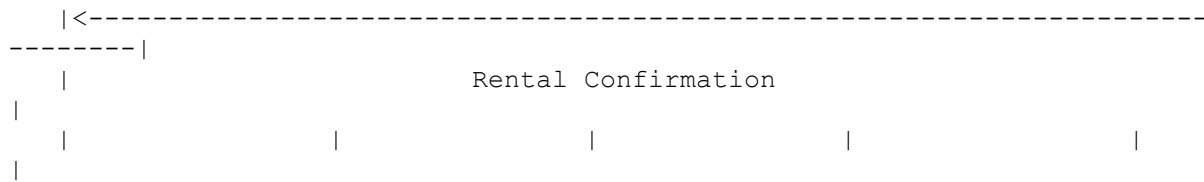
b) Message Sequence

1. Customer searches for vehicle.
2. Rental System requests available vehicles.
3. Vehicle Database returns available vehicles.
4. Rental System displays vehicles.
5. Customer selects vehicle.
6. Rental System sends booking request.
7. Booking System checks vehicle availability.
8. Booking System creates booking.
9. Customer makes payment.
10. Payment Gateway processes payment.

11. Payment Gateway returns payment confirmation.
12. Booking System confirms rental.
13. Notification Service sends rental confirmation.
14. Customer receives confirmation.

c) Sequence Diagram





Conclusion

The Sequence Diagram provides a complete time-ordered interaction between the customer and system components. It ensures that vehicle availability is checked before booking and payment confirmation is received before final rental confirmation.

4. State Diagram Construction – Traffic Signal Control System

Problem Analysis

The Traffic Signal Control System controls normal traffic signals and must also handle emergency vehicles and maintenance activities.

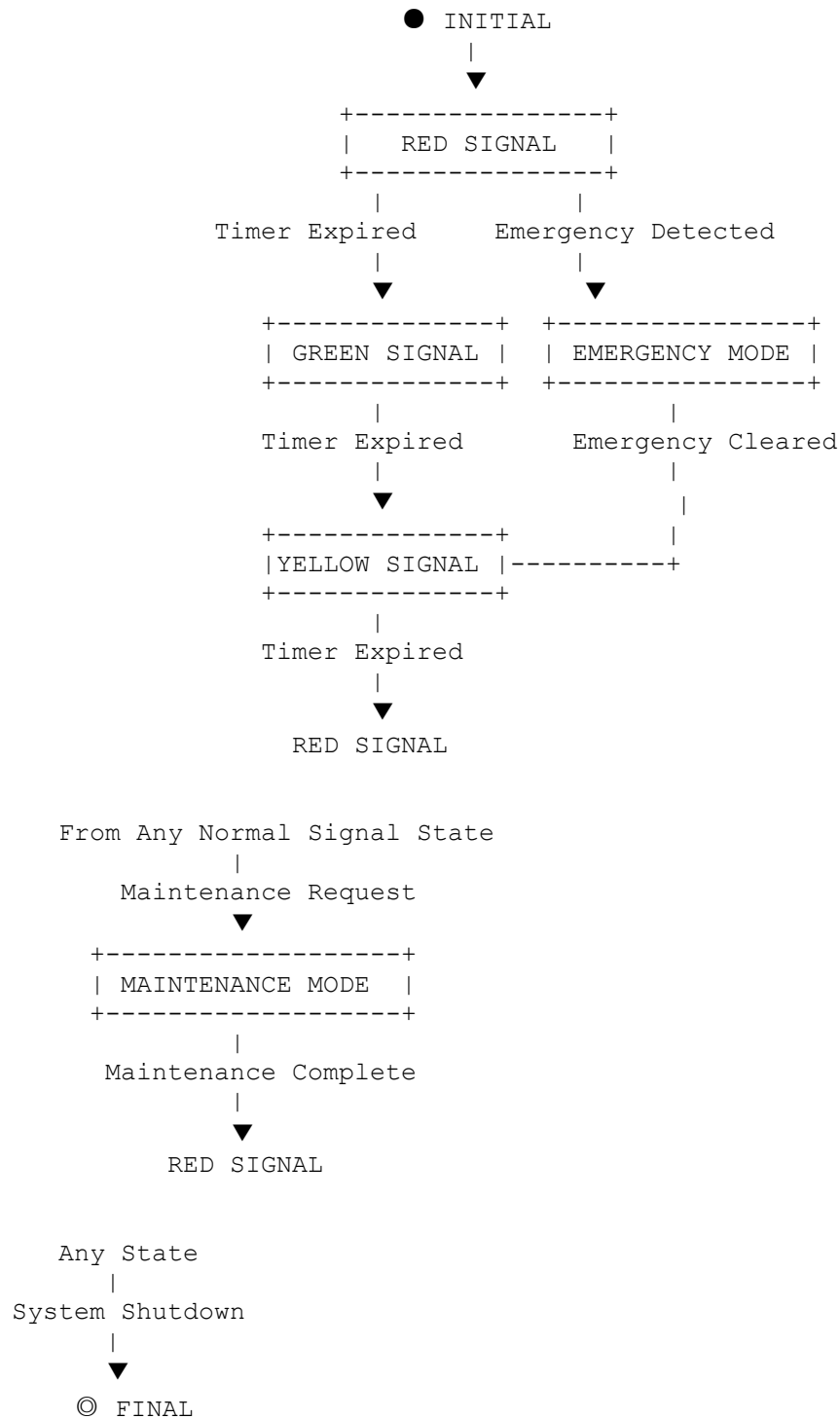
a) Valid States

1. **Initial**
2. **Red Signal**
3. **Green Signal**
4. **Yellow Signal**
5. **Emergency Mode**
6. **Maintenance Mode**
7. **Final/Shutdown**

b) Events and Conditions

- Timer expires
- Green time completed
- Yellow time completed
- Emergency detected
- Emergency cleared
- Maintenance request
- Maintenance completed
- System shutdown

c) State Transition Diagram



Detailed Transitions

Current State	Event	Next State
Red	Timer expires	Green
Green	Timer expires	Yellow

Current State	Event	Next State
Yellow	Timer expires	Red
Red/Green/Yellow	Emergency detected	Emergency Mode
Emergency Mode	Emergency cleared	Red
Any normal state	Maintenance request	Maintenance Mode
Maintenance Mode	Maintenance completed	Red
Any state	Shutdown	Final

Conclusion

The State Diagram clearly models the behavior of the traffic signal system. In normal operation, the signal follows **Red** → **Green** → **Yellow** → **Red**. Emergency detection moves the system to Emergency Mode, while maintenance requests move it to Maintenance Mode.

5. Component Diagram Design – Smart Healthcare Portal

Problem Analysis

The Smart Healthcare Portal consists of several independent but interconnected software components. The main goal is to clearly show how patient information, appointments, medical records, billing, and notifications interact.

a) Major Components

1. Patient Management
2. Appointment Scheduling
3. Medical Records
4. Billing
5. Notification Service
6. Authentication Service
7. Database

b) Interfaces

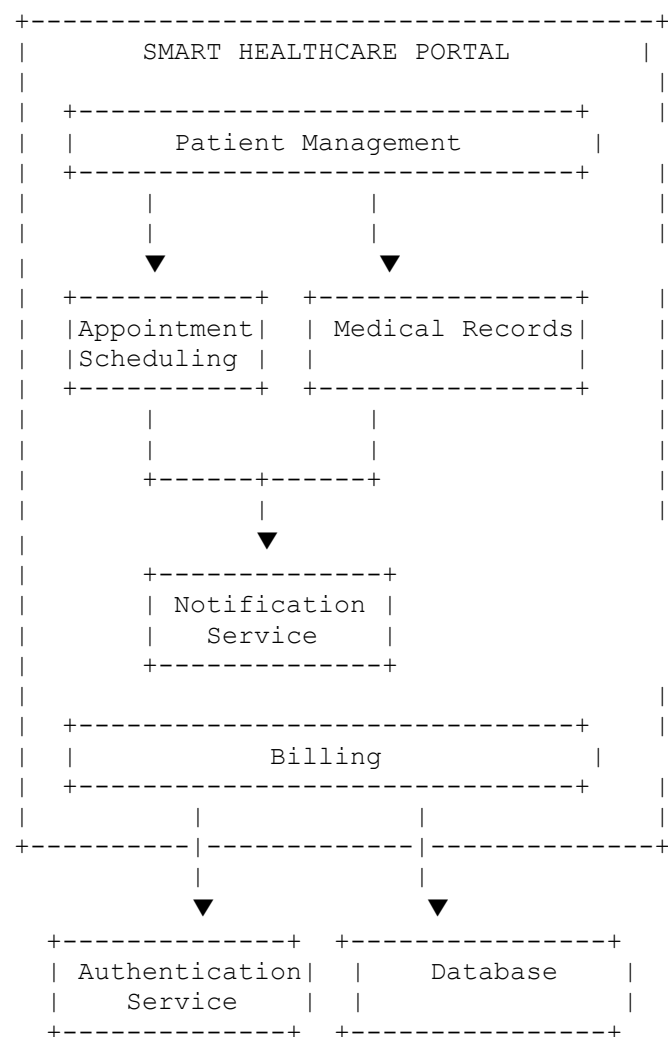
- Patient Management provides **Patient Information Interface**
- Appointment Scheduling provides **Appointment Interface**
- Medical Records provides **Medical Record Interface**
- Billing provides **Billing Interface**

- Notification Service provides **Notification Interface**
- Authentication Service provides **Authentication Interface**

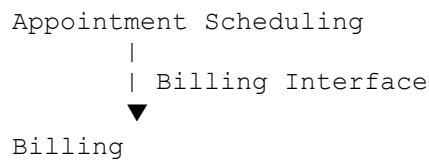
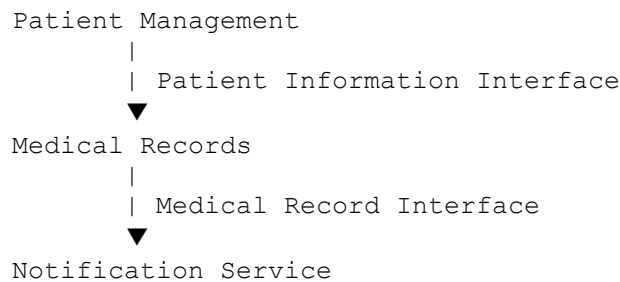
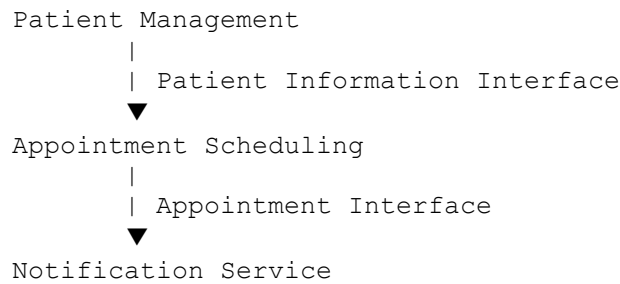
c) Dependencies

- Patient Management → Authentication
- Appointment Scheduling → Patient Management
- Medical Records → Patient Management
- Appointment Scheduling → Notification Service
- Billing → Patient Management
- Billing → Appointment Scheduling
- Medical Records → Notification Service
- All modules → Database

d) Component Diagram



Interface Relationships



Conclusion

The Component Diagram divides the Smart Healthcare Portal into modular components. The modules communicate through defined interfaces and depend on common services such as authentication, notification, and database storage.

6. Deployment Diagram Design – Smart City Waste Management System

Problem Analysis

The Smart City Waste Management System uses IoT sensors installed in waste bins to collect information such as fill level, temperature, and possible fire conditions. Edge devices process sensor data before sending it to cloud servers. City authorities monitor the system through web dashboards.

a) Hardware Nodes

1. Smart Waste Bin
2. IoT Sensor
3. Edge Gateway
4. Cloud Server
5. Cloud Database
6. Monitoring Dashboard
7. Administrator Computer/Mobile Device

b) Artifacts

Smart Waste Bin

- Fill Level Sensor
- Temperature Sensor
- IoT Sensor Software

Edge Gateway

- Data Collection Service
- Data Filtering Service
- Communication Service

Cloud Server

- Waste Management Application
- Route Optimization Module
- Alert Management Module
- Analytics Module

Cloud Database

- Waste Bin Data
- Collection History
- Sensor Data
- Alert Records

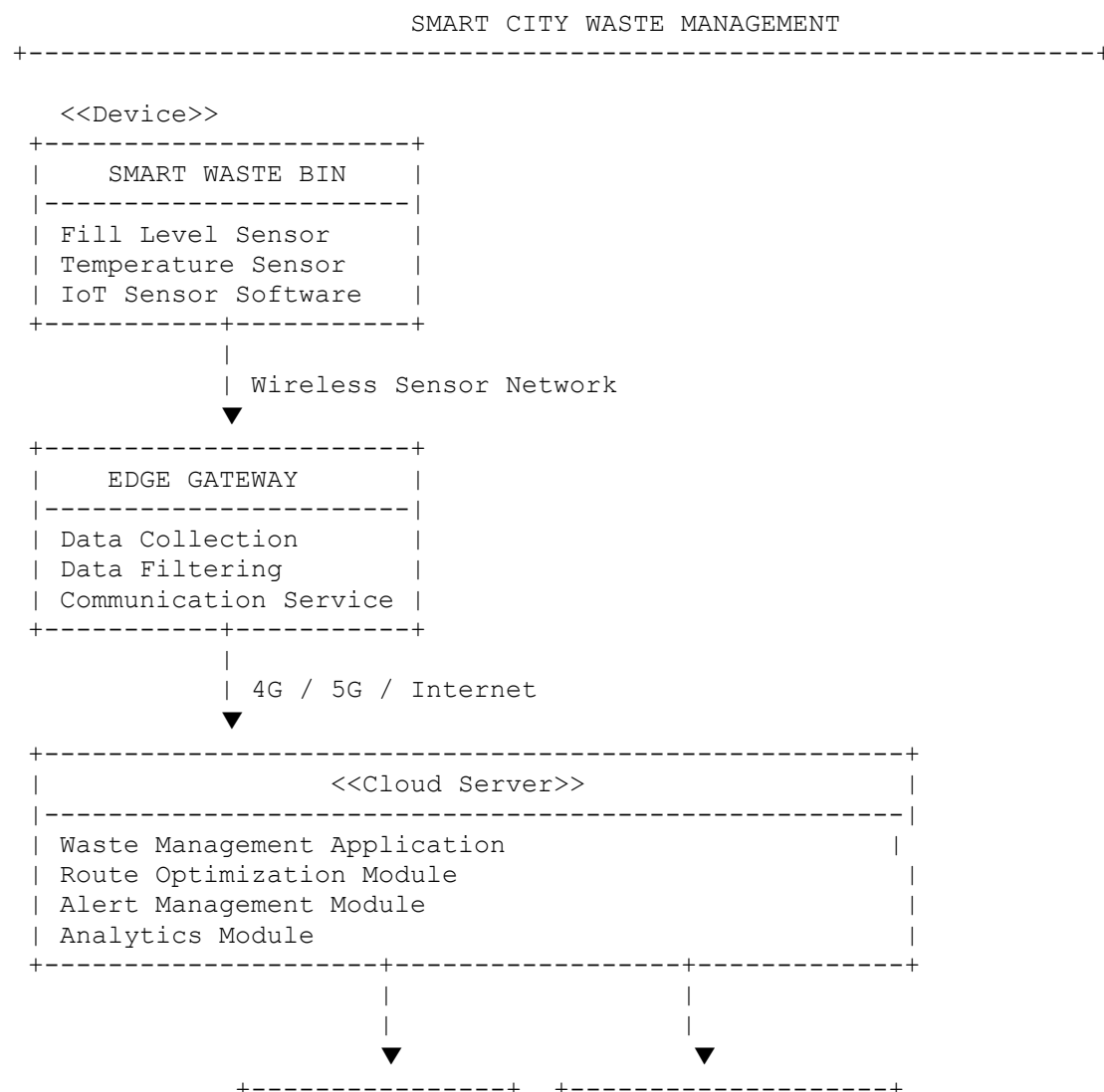
Monitoring Dashboard

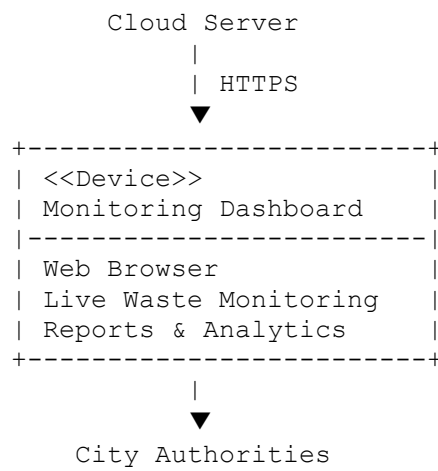
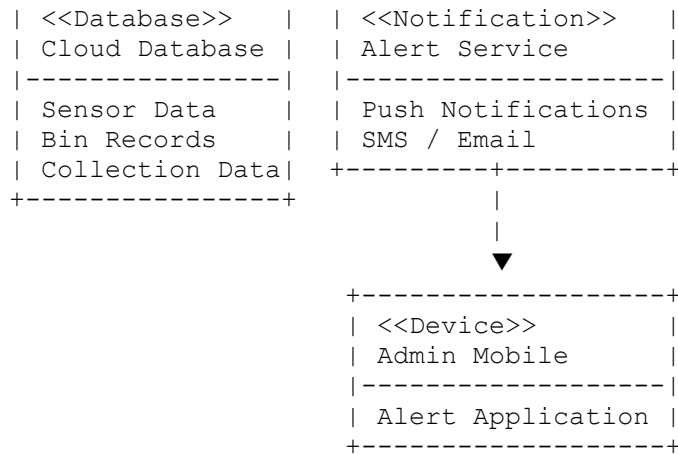
- Web Dashboard
- Reports
- Live Monitoring Interface

c) Communication Links

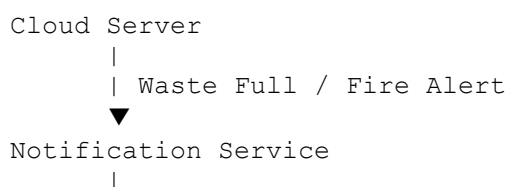
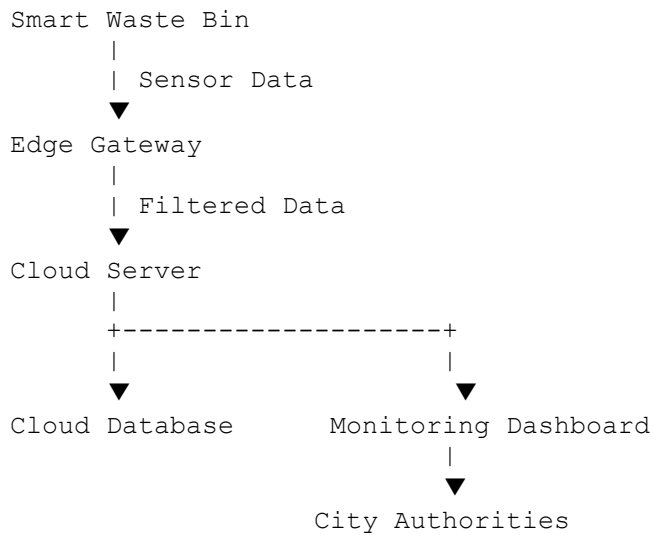
- IoT Sensors → Edge Gateway: Wireless Sensor Network
- Edge Gateway → Cloud Server: Internet / 4G / 5G
- Cloud Server → Cloud Database: Secure Internal Network
- Cloud Server → Dashboard: HTTPS
- Cloud Server → Mobile Device: Internet / Push Notification

d) Detailed Deployment Diagram





Complete Data Flow



▼
Admin Mobile Device

Conclusion

The Deployment Diagram represents the complete physical architecture of the Smart City Waste Management System. IoT sensors collect waste-bin data, edge gateways filter and transmit the data, and cloud servers store and analyze it. Monitoring dashboards provide real-time information to city authorities, while the notification service sends alerts when bins are full or abnormal conditions are detected.